

SANYAM KAUL

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EDUCATION

M.Tech Computer Science & Engineering

*IIT Hyderabad
2025 CGPA: 8.90*

B.Tech Computer Science & Engineering (Networks & Security)

*VIT-AP University, Amaravati
2021 CGPA: 8.90*

EXPERIENCE

Data Engineer

*Netradyne
July 2025 - Present*

- Designed and implemented AWS Monitoring Dashboard for infrastructure cost optimization, processing real-time metrics from EC2 instances, ASGs, and CloudWatch APIs to enable data-driven decision making and reduce infrastructure costs
- Built and optimized data pipelines using Python for Kubernetes EKS logging infrastructure, integrating rsyslog sidecar containers for real-time log streaming and forwarding to centralized monitoring systems
- Developed ETL workflows integrating Elasticsearch for log aggregation and retrieval, improving debugging efficiency for CI/CD pipelines by 40% through structured data processing and automated log analysis
- Engineered scalable reprocessing pipelines with TensorRT support (v8, v10), optimizing Docker-based compilation workflows and implementing modular, maintainable code architecture with backward compatibility
- Implemented data processing solutions for FSx-based network storage, reducing overall storage costs through efficient data lake architecture and optimized data access patterns

Tech Analyst (SDE-1)

*BlackRock
July 2021 - June 2023*

- Processed complex financial datasets from multiple sources including banks and payment processors, building Self-Service Financial Information Reporting Tool that reduced report setup time from 1.5 months to 2-3 days through automated data transformation pipelines
- Engineered scalable data processing applications using Python and Spring Boot, implementing intricate business rules and financial calculations with 85%+ test coverage, ensuring data integrity across different formats and structures
- Built event-driven CI/CD pipelines on Azure DevOps for 5 business-critical servers, achieving 50% automation in performance and integration testing through automated data validation workflows and strict SLA compliance
- Designed and implemented REST and GraphQL API integrations for data standardization, enabling seamless data exchange across 50+ business, development, and QA teams
- Developed real-time data processing solutions for financial metrics, ensuring accurate transaction processing, settlement, and reconciliation workflows with time-bound processing requirements

Tech Intern

*BlackRock
January 2021 - July 2021*

- Built real-time data visualization dashboard using Golang and Angular, processing CI/CD pipeline metrics and failure statistics from Azure DevOps APIs
- Implemented backend service for data streaming, fetching and processing CI/CD metrics to enable proactive monitoring and alerting

PROJECTS

HyperLedger Fabric Blockchain Optimization - M.Tech Thesis

*arXiv:2509.07425
IIT Hyderabad 2024-2025*

- Optimized core architecture of HyperLedger Fabric distributed blockchain system to reduce transaction failure rate, improving overall system efficiency and reliability
- Designed and implemented distributed system optimizations focusing on transaction processing, consensus mechanisms, and network architecture for enhanced performance
- Presented research findings at Indosys 2025 Conference, demonstrating expertise in building and optimizing complex distributed systems from scratch
- Published research paper on blockchain architecture improvements, showcasing deep understanding of distributed systems, performance tuning, and system optimization

WineSecretHK - RAG-based AI Food & Wine Pairing System

rag.winesecret.hk.com/docs

- Built RAG-based AI system with HyDE-inspired algorithm for flavor profile matching using vector similarity search; algorithm submitted for patent

- Implemented data pipeline with FastAPI backend, AWS RDS for structured data storage, and ChromaDB vector database for efficient similarity search and retrieval
- Designed event-driven architecture with webhook integrations for Stripe payments and Eats365 ordering platform, ensuring real-time data synchronization and transaction processing

TECHNICAL SKILLS

- **Programming:** Java, Python, TypeScript, JavaScript, Golang, C++; Strong OOPS concepts, modular and structured code design, excellent problem-solving skills
- **Big Data & Streaming:** Real-time data processing, Event-driven architectures, Distributed data processing
- **Cloud Platforms:** AWS (EC2, EKS, S3, FSx, CloudWatch, Cost Explorer APIs, RDS), Azure DevOps
- **Data Warehousing & Modeling:** AWS RDS, MySQL, MongoDB, Sybase, ChromaDB, S3 data lakes, Google Cloud Storage
- **ETL & Orchestration:** CI/CD pipelines, Jenkins, Data transformation workflows, Automated data processing
- **Distributed Systems:** HyperLedger Fabric, Blockchain architecture, Consensus mechanisms, System optimization, Performance tuning
- **DevOps & Tools:** Docker, Kubernetes, Git, Jira, Confluence, Elasticsearch, REST APIs, GraphQL, Version control

ACHIEVEMENTS

- 4/5 star rating in BlackRock annual employee ratings (2021, 2022)
- Published research paper on HyperLedger Fabric blockchain optimization (arXiv:2509.07425); Presented at Indosys 2025 Conference
- 1st position in IITH ML Hackathon; 15th position (out of 100+ teams) in IITH Deep Learning Hackathon
- 1st position in Object Oriented Programming course (100+ students); 450+ problems solved on LeetCode